Alex Siegel

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SUMMARY

I'm a postdoctoral scholar with over 10 years of experience applying biophysical and biochemical techniques to study the mechanisms of proteins. My research, published in *Science Advances*, *Nature Plants*, *PNAS*, among other journals, provides a detailed mechanistic understanding of how protein chaperones prevent the aggregation of their misfolding clients and offers a platform to build custom chaperones to combat aggregation-related diseases.

EDUCATION

University of California, Berkeley — Berkeley, CA
Ph.D. in Biophysics with specialization in Chemical Biology — June 2016
Graduate advisor – Prof. David E. Wemmer, Chemistry
Thesis: Mechanisms of σ54 bacterial transcription activation

California Institute of Technology — Pasadena, CA
B.S. in Biology and Chemistry, double major with honors — June 2008

RESEARCH EXPERIENCE

Postdoctoral Scholar in Chemistry at Caltech — Pasadena, CA
Research with Prof. Shu-ou Shan — 2017-present

- Used crosslinking-coupled mass spectrometry to map the interactions between a critical plant chaperone, cpSRP43, and a class of membrane protein LHCPs that form the photosynthesis machinery (McAvoy et al, *JBC*, 2018).
- Identified a new class of cpSRP43 clients involved in chlorophyll synthesis (Wang et al, *PNAS*, 2018) and characterized its novel protection mechanism under heat stress (Ji*, Siegel* et al, *Nature Plants*, 2021).
- Discovered a unique chaperone activation mechanism of cpSRP43 triggered by a disorder-to-order transition (Siegel*, McAvoy* et al, *JMB*, 2020) and showed how this switches its activity between membrane protein LHCPs and globular protein clients (Siegel et al, In Revision at *Science Advances*, 2025) or to non-native amyloid-forming clients (Prov. Patent ID: CIT-9049-P).
- Solved the cryo-EM structure of human mitochondrial ClpB homolog, Skd3, revealing its two roles as a disaggregase chaperone and as a chaperonin-like folding cage (Gupta*, Lentzsch*, Siegel*, et al, *Science Advances*, 2023).

Graduate Researcher in Biophysics at UC Berkeley — Berkeley, CA
Research with Prof. David Wemmer — 2009-2016

- Uncovered how transcriptional activators initiate transcription threading the N-terminus of an RNA Polymerase subunit through their central pore to apply the force needed to melt DNA (Siegel et al, *JMB*, 2016).
- Used molecular tweezers to understand force-dependent activation of RNA polymerase (Siegel, dissertation).

PUBLICATIONS

*Indicates equal contribution first author publications.

Siegel A, Kroon G, Zhao CQ, Wang P, Wright PE, and Shan SO. Switchable client specificity in a dual functional chaperone coordinates light harvesting complex biogenesis. In Revision at *Science Advances* (2025).

Siegel A*, Singh J*, Qin PZ & Shan SO EPR Studies of Chaperone Interactions and Dynamics. Biophysics of Molecular Chaperones (eds. Hiller, S., Liu, M. & He, L.) 242–277. *Royal Society of Chemistry* (2023).

Gupta, A*, Lentzsch AM*, **Siegel A***, Yu Z, Chio US, Cheng Y, & Shan SO. Dodecamer assembly of a metazoan AAA+ chaperone couples substrate extraction to refolding. *Science Advances*. 9(19), eadf5336 (2023).

Ji S*, **Siegel A***, Shan, S., Grimm, B. & Wang, P. Chloroplast SRP43 autonomously protects chlorophyll biosynthesis proteins against heat shock. *Nature Plants*. 7, 1420-1432 (2021).

Siegel A*, McAvoy C*, Lam V, Liang FC, Kroon G, Miaou E, Griffin P, Wright P, Shan SO. A Disorder-to-Order Transition Activates an ATP-Independent Membrane Protein Chaperone. *Journal of Molecular Biology*. 432, 166708 (2020).

McAvoy C, **Siegel A**, Piszkiewicz S, Miaou E, Yu M, Nguyen T, Moradian A, Sweredoski MJ, Hess S, Shan SO. Two Distinct Sites of client protein interaction with the chaperone cpSRP43. *Journal of Biological Chemistry*. RA118, (2018).

Wang P, Liang FC, Wittmann D, **Siegel A**, Shan SO, Grimm B. Chloroplast SRP43 acts as a chaperone for glutamyl-tRNA reductase, the rate-limiting enzyme in tetrapyrrole biosynthesis. *Proceedings of the National Academy of Sciences*. 115(15), E3588-96 (2018).

Siegel AR, Wemmer DE. Role of the σ54 Activator Interacting Domain in Bacterial Transcription Initiation. *Journal of Molecular Biology*. 428(23), 4669-85 (2016).

Siegel AR, Mechanisms of σ54 bacterial transcription activation. Doctoral Dissertation University of California, Berkeley. Publication No. 10192026. ProQuest Dissertations Publishing (2016).

PATENTS

Patent ID: CIT-9049-P. Provisional patent application for use of chaperones against neurodegenerative diseases.	Filed in 2023

PRESENTATIONS

*Oral presentation, †Poster presentation ‡Session chair

● 33rd Western Photosynthesis Conference*†‡	Biosphere 2, AZ (2024)
● ASBMB annual meeting in Experimental Biology Spotlight Session*†‡	Philadelphia, PA (2022)
● ASBMB Science in a Flash*	Virtual Meeting (2022)
● Center for Molecular and Cellular Medicine Biochemistry Seminar Series**	Caltech (2021 & 2021)
● EMBO workshop: New challenges in protein translocation across membranes*	Virtual Meeting (2021)
● Gordon Research Conference in Computational Aspects of Biomolecular NMR†	Lucca, Italy (2015)
● Structure Supergroup Biophysics Seminar Series**	UC Berkeley (2012 & 2014)
● Biophysics Annual Retreat††	UC Berkeley (2010 & 2014)

TEACHING AND MENTORING

Primary research mentor for 20 students including:
1 postdoc (6 mo.), 7 graduate (31 mo.), 6 undergraduate (65 mo.), 6 high school (142 mo.)

Courses taught as *Lead Instructor of Record, †Discussion Section TA, ‡Lab Course TA

● Advanced Molecular Biology Lab‡†	UC Berkeley (2016)
● Biophysics module on protein NMR spectroscopy*	UC Berkeley (2012)
● Introductory Chemistry Lab‡†	UC Berkeley (2014)
● Biophysical Chemistry†	UC Berkeley (2010)
● Introductory Biology†	Caltech (2007 & 2008)

SKILLS

*Experimental or computational skills with proficiency (instruments/software/methods used)

NMR Spectroscopy Graduate and postgraduate research experience using solution-state NMR Spectroscopy to study protein structure and dynamics. ● Isotopically labeled sample preparation (M9 media growth) ● Spectra collection (Bruker Topspin) ● Writing pulse programs (Bruker Topspin) ● Data analysis (NMRpipe, NMRview, CARA, MestreNova) ● Spectrometer maintenance (Bruker 600-900 MHz, nitrogen & helium fills, basic maintenance). ● Isotopic labeling (^{15}N, ^{13}C ILV, site-directed spin labeling with MTSL & MMTS, ^{19}F NMR phenylalanine & BTFA labeling) ● Multiple dimension collection and analysis (Diffusion ordered spectroscopy, Assignments via HNCA etc., Structure determination by NOE) ● Paramagnetic relaxation enhancement ● Non-uniform sampling

Biophysics Techniques ● X-Ray Crystallography (LBNL synchrotron independent use) ● Mass Spectrometry (Orbitrap Elite, Q-Exactive) ● Electron Paramagnetic Resonance (Bruker CW-EPR spectrometer) ● Cryo Electron Microscopy (Titan Krios) ● Small Angle X-ray Scattering ● Fluorescence Anisotropy and Bulk FRET (Horiba Fluorolog) ● Single molecule FRET (ALEX, Hamamatsu Photonics picosecond streak camera) ● Fluorescence lifetime measurements (Horiba TCSPC) ● Mass Photometry (Refeyn) ● Circular Dichroism (Aviv model 410) ● Molecular Tweezers

Molecular Biology Techniques ● Cloning (site-directed mutagenesis, Gibson Assembly) ● Protein expression (E. coli) ● Protein purification (FPLC/Chromatography) ● Immunoassays (western blotting and quantification, immunoprecipitation) ● Amber suppression (Schultz method) ● In vitro translation (PURE and self-prepared systems) ● PCR/qPCR

Biochemistry Techniques ● Light scattering (Beckman DU series) ● Luciferase folding assays (SpectraMax iD5 plate reader) ● ATPase assays ● UV crosslinking (Bpa) ● Chemical crosslinking (BMOE, BMH, DSS, DSSO, DHSO) ● Fluorescence labeling (AlexaFluor/BODIPY maleimides) ● EPR and NMR labeling (MTSL) ● Chromatography (Agilent 1100 series HPLC).

Software ● Structure visualization (ChimeraX, Pymol) ● CryoEM software (CryoSPARC, Relion) ● Mass spectrometry (ProteinProspector, MeroX) ● Protein structure determination (PHENIX, Coot) ● Structure computation (Gromacs, AlphaFold 2) ● Bioinformatics (BioPython) ● Data analysis (R, Matlab, Graphpad Prism) ● Coding languages (Python)